MATH317 · NUMERICAL METHODS

Homework 1 Solutions

Errors · Taylor · Horner

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01 Question 1: finding x from the relative error

A real number x is represented approximately by 0.6032, and the relative error is at most 0.1%. What is x?

The relative error is at most `0.1% = 0.001`. Write the condition:

$$\frac{|x - 0.6032|}{|x|} \leq 0.001$$

$$-0.001x \leq x - 0.6032 \leq 0.001x$$

$$0.999x \leq 0.6032 \quad \text{and} \quad 0.6032 \leq 1.001x$$

$$\frac{0.6032}{1.001} \leq x \leq \frac{0.6032}{0.999}$$

$$\mathbf{0.6026 \leq x \leq 0.6038}$$

So `x` is not a single number; it is any value inside this interval.

02 Question 2: rounding error

What is the relative error involved in rounding 4.9997 to 5.000?

Exact = 4.9997 Approximate = 5.000

$$\text{Relative error} = \frac{|4.9997 - 5.000|}{4.9997} = \frac{0.0003}{4.9997}$$

$$\approx 6.0 \times 10^{-5} \quad (0.006\ \%)$$

03 Question 3: the series for $(1 + x)^n$

Task: derive the Maclaurin series of $(1 + x)^n$, apply it for $n = 2$, $n = 3$ and $n = \frac{1}{2}$, then compute $\sqrt{1.0001}$ to 15 places.

1 Differentiate and put $x = 0$

$f(x) = (1+x)^n$	$f(0) = 1$
$f'(x) = n(1+x)^{n-1}$	$f'(0) = n$
$f''(x) = n(n-1)(1+x)^{n-2}$	$f''(0) = n(n-1)$
$f'''(x) = n(n-1)(n-2)(1+x)^{n-3}$	$f'''(0) = n(n-1)(n-2)$

2 Substitute into the Maclaurin formula

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

3 Special cases

$n = 2$: $(1+x)^2 = 1 + 2x + x^2$

$n = 3$: $(1+x)^3 = 1 + 3x + 3x^2 + x^3$

$n = \frac{1}{2}$: $\sqrt{(1+x)} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{5x^4}{128} + \ldots$

4 Compute $\sqrt{1.0001}$ with $x = 10^{-4}$

$\frac{x}{2} = 0.00005$

$\frac{x^2}{8} = 0.00000000125$

$\frac{x^3}{16} = 0.0000000000000625$

$\frac{5x^4}{128} \approx 3.9 \times 10^{-18}$ (too small to matter)

$\sqrt{1.0001} \approx 1 + 0.00005 - 0.00000000125 + 0.0000000000000625$

$\sqrt{1.0001} \approx 1.000049998750062$

04 Question 4: how many terms?

Task: the number of terms of the series for $\ln(1 + x)$ needed to compute $\ln 1.1$ with error below $\frac{1}{2} \times 10^{-8}$.

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ $x = 0.1$

The series is alternating with decreasing terms, so the error is at most the first term left out.

After n terms: $\text{error} \leq \frac{(0.1)^{n+1}}{n+1}$

Need: $\frac{(0.1)^{n+1}}{n+1} < 5 \times 10^{-9}$

n = 6: $\frac{10^{-7}}{7} = 1.43 \times 10^{-8}$ (not enough)

n = 7: $\frac{10^{-8}}{8} = 1.25 \times 10^{-9}$ (enough)

7 terms

05 Question 5: Taylor in terms of h

Task: write `f(2 + h)` for `f(x) = x³ − 2x² + 4x − 1`.

$$f(2+h) = f(2) + f'(2)h + \frac{f''(2)}{2!}h^2 + \frac{f'''(2)}{3!}h^3$$

$f(x) = x^3 - 2x^2 + 4x - 1$ $\quad$ $f(2) = 8 - 8 + 8 - 1 = 7$

$f'(x) = 3x^2 - 4x + 4$ $\quad$ $f'(2) = 12 - 8 + 4 = 8$

$f''(x) = 6x - 4$ $\quad$ $f''(2) = 8$

$f'''(x) = 6$ $\quad$ $f'''(2) = 6$

$$f(2+h) = 7 + 8h + (\frac{8}{2})h^2 + (\frac{6}{6})h^3$$

$f(2+h) = 7 + 8h + 4h^2 + h^3$

Check with `h = 1`: `f(3) = 27 − 18 + 12 − 1 = 20` and `7 + 8 + 4 + 1 = 20`.

06 Question 6: complete Horner

Task: the Taylor expansion of this polynomial about `r = 2`.

$$p(x) = 3x^5 - 2x^4 + 15x^3 + 13x^2 - 12x - 5$$

	3	−2	15	13	−12	−5	
2 |		6	8	46	118	212	
	3	4	23	59	106 |	207	c0 = 207
		6	20	86	290		
	3	10	43	145 |	396		c1 = 396
		6	32	150			
	3	16	75 |	295			c2 = 295
		6	44				
	3	22 |	119				c3 = 119
		6					
	3 |	28					c4 = 28
	3						c5 = 3

$$p(x) = 3(x-2)^5 + 28(x-2)^4 + 119(x-2)^3 + 295(x-2)^2 + 396(x-2) + 207$$

Check: $p(2) = 96 - 32 + 120 + 52 - 24 - 5 = 207$.

The same result with the pseudocode

```
for k = 0 to n-1
    for j = n-1 downto k
        a_j ← a_j + r · a_{j+1}
```

	a_0	a_1	a_2	a_3	a_4	a_5
start	−5	−12	13	15	−2	3
k = 0	207	106	59	23	4	3
k = 1	207	396	145	43	10	3
k = 2	207	396	295	75	16	3
k = 3	207	396	295	119	22	3
k = 4	207	396	295	119	28	3

At the end a_k is the coefficient of $(x - 2)^k$, the same answer as the table.

07 Question 7: Taylor about c = 3

$$f(x) = x^4 - 4x^3 + 7x^2 - 5x - 2$$

	1	−4	7	−5	−2	
3 |		3	−3	12	21	
	1	−1	4	7 |	19	c0 = 19
		3	6	30		
	1	2	10 |	37		c1 = 37
		3	15			
	1	5 |	25			c2 = 25
		3				
	1 |	8				c3 = 8
	1					c4 = 1

$$\mathbf{f(x) = (x-3)^4 + 8(x-3)^3 + 25(x-3)^2 + 37(x-3) + 19}$$

Check: `f(3) = 81 − 108 + 63 − 15 − 2 = 19`.

Important for the midterm

Questions 6 and 7 follow the same pattern as the Horner question that appeared in three midterms.

The wording of questions 3 to 7 is paraphrased from my earlier solution of this homework, not copied word for word from the sheet.